Dr. Francesca Dominici

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Website ORCID Google Scholar

Education & Training

- B.S. in Statistics, University La Sapienza, Rome, Italy, 110/110 (cum laude), 1993
- Ph.D. in Statistics, University of Padua, Italy, 1997
- Master of Public Health (Honorary), Harvard T.H. Chan School of Public Health, Harvard University, 2009

Areas of Research

Bayesian statistics, causal inference, artificial intelligence/machine learning, climate change, environmental epidemiology, health policy, life science

Professional Experience

- Founding Director, Harvard Data Science Initiative, Harvard University, 2017–Present
- Clarence James Gamble Professor of Biostatistics, Population Health, and Data Science; Faculty Affiliate, Dept. of Environmental Health, Harvard T.H. Chan School of Public Health, 2009–Present
- Contributor, Forbes 2024-present
- Senior Associate Dean for Research, Harvard T.H. Chan School of Public Health, 2013–2017
- Associate Dean of Information Technology, Harvard T.H. Chan School of Public Health, 2011–2013
- Professor, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, 2007–2009
- Associate Professor, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, 2002–2007
- Joint Appointment, Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health, 2001–2009
- Assistant Professor, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, 1999–2002
- Visiting Assistant Professor, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, 1997–1999
- Visiting Scholar, Institute of Statistics and Decision Science, Duke University, 1995–1997

Academic Leadership

- Member, Presidential Nomination MIT President Visiting Committee, MIT Institute for Data, Systems, and Society, 2024–Present
- Faculty Associate, Harvard Global Health Institute, 2024–Present
- Faculty Associate, Harvard Center of International Development, 2024–Present
- Steering Committee Member, Kempner Institute for the Study of Natural and Artificial Intelligence, Harvard University
- Steering Committee Member, #metoo Digital Media Project, Radcliffe Institute, Harvard University
- Faculty Associate, Salata Institute for Climate and Sustainability, Harvard University, 2021–Present
- Steering Committee Member, Harvard Library, 2020–Present
- Steering Committee Member, Institute of Quantitative Social Science, Harvard University, 2015–Present
- Director, National Studies on Air Pollution and Health (NSAPH) Group, Harvard T.H. Chan School of Public Health, 2010–Present
- Program Co-Director, Environmental Stats Training Grant, Harvard T.H. Chan School of Public Health, 2018–2022

University Service

- University Council on Research Computing (Chair: James Waldo, Harvard CTO), 2011–Present
- Harvard Academic Computing (Chair: Anne Margulies, Harvard CIO), 2011–Present
- Professorial Review Committee (Chair: James Kasser, Harvard Medical School), 2015–2016
- Big Data Planning Committee (Chair: Richard McCullough, Harvard VP of Research), 2014–2017
- Member, Ad Hoc Committee to Advise the Dean for Tenured Appointments, Harvard Medical School, 2012
- Member, Ad Hoc Committee to Advise the Provost for Tenured Appointments, Harvard School of Public Health, 2012
- Chair, Ad Hoc Committee to Advise the Provost for Tenured Appointments, Harvard School of Public Health, 2011

Press Coverage (Selected)

- "Most New England States Don't Track Heat-Related Deaths. Experts Say That's a Problem." WBUR, Aug 31, 2023
- "Cleaner Air Helps Everyone. It Helps Black Communities a Lot." New York Times, Mar 23, 2023
- "Even Low Levels of Soot Can Be Deadly to Older People, Research Finds." New York Times, Jan 26, 2022
- "Wildfire Smoke Linked to Higher COVID-19 Death Rates." National Geographic, Aug 13, 2021
- "Air Pollution Kills Millions Every Year, Like a 'Pandemic in Slow Motion'." National Geographic, Mar 16, 2021
- "Pollution Made COVID-19 Worse. Now, Lockdowns are Clearing the Air." National Geographic, Apr 8, 2020
- "Covid-19 Death Rate Rises in Counties with High Air Pollution, Study Says." CNN, Apr 7, 2020
- "U.S. Air Pollution Still Kills Thousands Every Year, Study Concludes." NPR, Jun 28, 2017

Professional Activities

Society Memberships and Leadership

- Elected Fellow, International Society of Bayesian Analysis
- Elected Fellow, Institute of Mathematical Statistics
- Elected Member, National Academy of Medicine
- Elected Fellow, American Statistical Association
- International Biometric Society
- International Society for Environmental Epidemiology
- Program Committee Member, Junior Researcher Workshop, International Biometric Society – ENAR Spring Meeting, Washington, DC, 2003
- Program Committee Member, International Biometric Society – ENAR Spring Meeting, Washington, DC, 2002
- Chair, Program Committee (Junior Researcher Workshop), International Biometric Society – ENAR Spring Meeting, Washington, DC, 2002

Participation on Advisory Panels

- Board Member Fineco Bank
- Advisory Board Member, *Harvard Data Science Review*, Jul 2019–Present
- Scientific Advisory Committee Member, Solutions for Energy, AiR, Climate, and Health Center (EPA Air, Climate, and Energy Center), Yale University, Aug 2016–Present
- Medical Advisory Board Member, RAPIDSOS Inc., Jan 2016–Present
- John B. Little International Scientific Advisory Board Member, Harvard T.H. Chan School of Public Health, Oct 2015–Present
- Scientific Advisory Board Member, Radiation Effects Research Foundation, Jun 2014–Jun 2022
- Chairperson, NIH Biostatistical Methods and Research Design Study Section (BMRD), Center for Scientific Review, Jul 2017–Jul 2018
- Member, Centre of Environmental Health Research Advisory Council, Public Health Foundation of India, May 2016–2019
- Member, Institute of Medicine Committee on Designing an Epidemiologic Study for Multiple Sclerosis and Other Neurologic Diseases in pre- and post-9/11 Gulf War Veterans, National Academy of Sciences, Jan 2015–Jan 2016
- National Advisory Committee Member, Statistical and Applied Mathematical Sciences Institute, Jan 2015–2019
- Research Committee Member, Health Effects Institute, Nov 2014–Nov 2022
- Ad Hoc Committee Member, JPB Environmental Health Fellows Program, Apr 2014–2016
- Member, Institute of Medicine Committee: Research on Health Effects of Low-Level Ionizing Radiation Exposure (AFRRI), National Academy of Sciences, Apr 2013–Jun 2014
- Review Panel Member, Advising Deans (NCSU CPMS/CALS/Graduate School), Dept. of Statistics, North Carolina State University, 2013
- Member, NIH BMRD Study Section, Center for Scientific Review, Jul 2012–2016
- Member, Institute of Medicine Committee on Gulf War and Health: Long-Term Effects of Blast Exposures, National Academy of Sciences, Jun 2012–Aug 2013
- Chair, External Review Committee, Dept. of Biostatistics, Mailman School of Public Health, Columbia University, May 2012
- External Advisory Board Member, Southern California Environmental Health Sciences Center, 2011–2017
- Member, Institute of Medicine Committee to Review the Federal Response to the Health Effects Associated with the Gulf of Mexico Oil Spill, National Academy of Sciences, Sep 2010–Apr 2011
- Member, Institute of Medicine Committee on Long-Term Health Consequences of Exposure to Burn Pits in Iraq and Afghanistan, National Academy of Sciences, Feb 2010–Apr 2011

- Member, Institute of Medicine Committee on Secondhand Smoke Exposure and Acute Coronary Events, National Academy of Sciences, Jan 2009–Aug 2009
- Member, Institute of Medicine Committee to Review ATSDR's Great Lakes Report, National Academy of Sciences, Apr 2008–Aug 2008
- Member, Institute of Medicine Committee on Making Best Use of the Agent Orange Exposure Reconstruction Model, National Academy of Sciences, 2007–2008
- Member, Institute of Medicine Committee on Gulf War and Health, National Academy of Sciences, 2005–2006
- External Advisory Board Member, MESA Air Pollution Study (PI: Joel Kaufman), Univ. of Washington, 2005–2015
- External Review Committee Member, NIH NICHD Division of Epidemiology, Statistics & Prevention Research, 2004
- Advisory Board Member, Eastern North American Region, International Biometric Society, 2003–2005
- Member, IOM Committee to Assess Potential Health Effects from Exposures to PAVE PAWS RF Energy, Cape Cod, MA, NAS, 2002–2005
- Member, Scientific Review Panel, Center for Environmental Statistics, EPA Study Section, 2001

Leadership and Diversity

- Chair, Committee for the Advancement of Women Faculty, Harvard T.H. Chan School of Public Health, Aug 2015–2017
- Attendee, Executive Program on Negotiation and Leadership, Harvard Law School Executive Education, May 18–20, 2015
- Panelist, "Lessons on Leadership: Defying Gender Stereotypes in the Workplace," Harvard T.H. Chan School of Public Health, Mar 2014
- Panelist, "Having it All: Weighting to Achieve Balance," ENAR, International Biometric Society, Mar 2014
- Participant, New Faculty Panel Discussion, Office of Faculty Development and Diversity, Harvard University, 2013
- Co-Chair, Committee for the Advancement of Women Faculty, Harvard T.H. Chan School of Public Health, 2012–Aug 2015
- Leader, Committee on the Concerns of Women Panel, "Imperfect Balance," Harvard University, 2012
- Author, "So Few Women Leaders," American Association of University Professors, 2009
- Diversity Recognition Award, President of Johns Hopkins University, 2009
- Co-Chair (with Linda P. Fried), University Committee on the Status of Women, Johns Hopkins University, 2005–2009

Consultations

- Litigation: *Kerr et al. vs. Aerojet General Corporation* (Neuroblastoma Incidence Analysis), Chino Hills, CA, 2005
- National Academy of Sciences Committee on Research Priorities for Airborne Particulate Matter, 2002
- Clean Air Act Advisory Committee, U.S. EPA, 2002
- Pacific Business Group on Health: The 1998 Asthma Outcome Survey, 1998
- Kaiser Permanente: Morbidity and Mortality Associated with Air Pollution among California Members, 1998
- Litigation: *Magaly Pinares et al. vs. United Technologies Corporation, Pratt & Whitney Group*, May 2018
- Litigation: *Boggs, Charles v. BP* (Deepwater Horizon, Case No. 16-cv-13476, The Downs Law Group), 2022–Present

Testimony

- Maryland State Committee on House Economic Matters: Health Effects of Particulate Matter and Ozone, 2005
- Maryland State Committee on Education, Health, and Environmental Affairs: Health Effects of Particulate Matter and Ozone, 2005

Editorial Activities

Editorial Board Activities

- Co-Editor in Chief, *Harvard Data Science Review*
- Senior Editor, Chapman & Hall/CRC Texts in Statistical Science Series
- Associate Editor, *Journal of the Royal Statistical Society*, 2011–2015
- Editor, *American Journal of Epidemiology* (Statistical Methodology), 2002–2007

Peer Review (Selected Journals)

- *American Journal of Epidemiology*; *Biometrics*; *Biostatistics*; *Environmetrics*; *Environmental Health Perspectives*; *Epidemiology*; *Journal of the American Statistical Association*; *Journal of the Royal Statistical Society*; *Journal of the American Medical Association*; *New England Journal of Medicine*; *Science*; *Statistics in Medicine*; *Nature*

Ad Hoc Review of Proposals

- NIH BMRD Study Section, CSR (Jul 1, 2012–Jun 30, 2018)
- U.S. Environmental Protection Agency
- Health Effects Institute, Boston, MA
- National Institutes of Health (NIH)
- National Institute of Environmental Health Sciences (NIEHS)
- Retirement Research Foundation
- Book proposals: John Wiley & Sons; Springer

Honors & Awards

Honors

- NIH WALS Award Lecture, 2026 The series features some of the biggest names in biomedical and behavioral research each season, and all speakers are nominated by the NIH community.
- Named one of the Top Italian Women Scientists in the Club of Fondazione Onda, 2026. Bestowed upon scientists with an H-index higher than 60.
- Named to *TIME* 100 Most Influential People in Global Health, 2024
- One of four scientists worldwide nominated by *Science* editors as a pioneer in data science and scientist to watch, 2024
- Top 50 most powerful women in Italy 2024
- NIEHS Key Paper of the Year: "Air Pollution and Mortality at the Intersection of Race and Social Class," *NEJM*, 2023
- NIEHS Key Paper of the Year: "Association of Tropical Cyclones with County-Level Mortality in the US," *JAMA*, 2022
- NIEHS Key Paper of the Year: "Air Pollution Exposure Disparities across US Population and Income Groups," *Nature*, 2022
- NIEHS Key Paper of the Year: "Tropical Cyclone Exposure is Associated with Increased Hospitalization Rates in Older Adults," *Nature Communications*, 2021
- NIEHS Key Paper of the Year: "Excess of COVID-19 Cases and Deaths due to Fine Particulate Matter Exposure during the 2020 Wildfires in the United States," *Science Advances*, 2021
- Elected Member, International Society of Mathematical Statistics, 2020
- Clarivate Highly Cited Researcher (top 1% worldwide), 2019
- Presidential Invited Address, International Biometric Society – ENAR, 2019
- Elected Member, National Academy of Medicine, 2018

- Recognized among top 10 Italian women scientists in biomedical sciences worldwide, 2016
- NIEHS Paper of the Year (Honorable Mention): "Cause-specific Risk of Hospital Admission Related to Heat in Older Adults," *JAMA* 2015
- Clarivate Highly Cited Researcher (top 1% worldwide), 2015
- Wierman Distinguished Lecture, Johns Hopkins University, 2015
- Best Environmental Epidemiology Papers (Honorable Mention): "Residential Exposure to Aircraft Noise and Hospital Admissions for Cardiovascular Diseases," *BMJ*, 2013
- Speaker, Mathematics for Planet Earth 2013 Simons Foundation Public Lecture Series (SAMSI), 2013
- Keynote Speaker, Environmental Health Day, University of Michigan, 2011
- Dean's Lecture, Bloomberg School of Public Health, Johns Hopkins University, 2007
- Fellow, Johns Hopkins Center of Excellence in Public Health Tracking (CDC), 2005
- Elected Fellow, American Statistical Association, 2005
- Keynote Speaker, Royal Statistical Society, London, 2002

Awards

- Vartziotis T, Kosteli R, Vartziotis E, Dasoulas G, Keckeisen M, Skianis K, Kotsopoulos S, Dominici F (2026). From tokens to watt-hours: analytical energy estimation for LLM inference on modern GPUs. arXiv [Preprint]. arXiv:2607.26571. doi:10.48550/arXiv.2607.26571 (Best Paper Award at Green-AI Workshop, Naples, 2026)
- Mosteller Statistician of the Year, Boston Chapter of the American Statistical Association (for exceptional contributions to statistics and service), 2022
- Star-Friedman Challenge Award for Promising Scientific Research, Harvard University (high-risk, high-impact research), 2020
- Karl E. Peace Award for Outstanding Statistical Contributions for the Betterment of Society, American Statistical Association, 2020
- Jerome Sacks Award for Cross-Disciplinary Research, American Statistical Association, 2020
- Janet L. Norwood Award for Outstanding Achievement by a Woman in the Statistical Sciences, University of Alabama, 2016
- Florence Nightingale David Award (COPSS and Caucus for Women in Statistics), 2015
- Diversity Recognition Award, Johns Hopkins University, 2009
- Myrto Lefkopoulou Distinguished Lectureship, Harvard University, 2007
- Gertrude Cox Award, Washington DC Chapter of the ASA and RTI International, 2007
- Mortimer Spiegelman Award, American Public Health Association (Statistics Section), 2006

- Young Investigator Award for Best Paper of the Year, ASA Section on Statistics in Epidemiology, 2001
- Walter A. Rosenblith New Investigator Award (3-year fellowship), Health Effects Institute, 1999–2002
- Student Award, International Biometric Society – ENAR, 1998
- Three-Year Fellowship & Award for Study and Research Abroad, Dept. of Statistics, University of Padua, Italy, 1994–1997

Journal Articles (*peer reviewed*)

1. **Dominici F**, Parmigiani G, Reckhow KH, Wolpert RL (1997). Combining Information from Related Regressions. *Journal of Agricultural, Biological, and Environmental Statistics* 2(3):294–312.
2. **Dominici F** (1997). Testing Simultaneous Hypotheses in Pharmaceutical Trials: A Bayesian Approach. *Journal of Biopharmaceutical Statistics* 8(2):283–297.
3. Zeger SL, **Dominici F**, Samet JM (1999). Harvesting-Resistant Estimates of Pollution Effects on Mortality. *Epidemiology* 10(2):171–175.
4. **Dominici F**, Parmigiani G, Wolpert RL, Hasselblad V (1999). Meta-analysis of Migraine Headache Treatments: Combining Information from Heterogeneous Designs. *Journal of the American Statistical Association* 94(445):16–28.
5. **Dominici F** (2000). Combining Contingency Tables with Missing Dimensions. *Biometrics* 56(2):546–553.
6. **Dominici F**, Parmigiani G, Clyde MA (2000). Conjugate Analysis of Multivariate Normal Data with Incomplete Observations. *Canadian Journal of Statistics* 28(3):533–550.
7. **Dominici F**, Zeger SL, Samet JM (2000). A Measurement Error Model for Time-Series Studies of Air Pollution and Mortality. *Biostatistics* 2:157–175.
8. Zeger S, Thomas D, **Dominici F**, Cohen AJ, Schwartz J, Dockery D, Samet JM (2000). Measurement Error in Time-series Studies of Air Pollution: Concepts and Consequences. *Environmental Health Perspectives* 108(5):419–426. PMID: 10811568; PMCID: PMC1638034; doi: 10.1289/ehp.00108419.
9. **Dominici F**, Samet JM, Zeger SL (2000). Combining Evidence on Air Pollution and Daily Mortality from the Twenty Largest US Cities: A Hierarchical Modeling Strategy (with discussion). *Journal of the Royal Statistical Society: Series A* 163(3):263–302. (Read at the Royal Statistical Society, London, Jan 10, 2000.)
10. Daniels M, **Dominici F**, Samet JM, Zeger SL (2000). Estimating PM10–Mortality Dose-Response Curves and Threshold Levels: An Analysis of Daily Time-Series for the 20 Largest US Cities (with discussion). *American Journal of Epidemiology* 152(5):397–412.
11. Samet JM, **Dominici F**, Curriero F, Coursac I, Zeger SL (2000). Fine Particulate Air Pollution and Mortality in 20 U.S. Cities: 1987–1994 (with discussion). *New England Journal of Medicine* 343(24):1742–1749.

12. **Dominici F**, Parmigiani G (2001). Bayesian Semiparametric Dose-Response Models: An Application to Toxicology. *Biometrics* 57(1):166–173.

13. **Dominici F** (2002). Invited Commentary on: Air Pollution and Health: What Can We Learn from a Hierarchical Approach? *American Journal of Epidemiology* 155(1):11–15.

14. **Dominici F**, Daniels M, Zeger SL, Samet JM (2002). Air Pollution and Mortality: Estimating Regional and National Dose-Response Relationships. *Journal of the American Statistical Association* 97(457):100–111.

15. Diette GB, Krishnan J, **Dominici F**, Haponik E, Skinner EA, Steinwachs D, Wu AW (2002). Asthma in the Elderly: Factors Associated with Hospitalization. *Archives of Internal Medicine* 162:1123–1132.

16. **Dominici F**, McDermott A, Zeger SL, Samet JM (2002). On the Use of Generalized Additive Models in Time-Series Studies of Air Pollution and Health. *American Journal of Epidemiology* 156(3):193–203.

17. **Dominici F**, McDermott A, Zeger SL, Samet JM (2003). National Maps of the Effects of Particulate Matter on Mortality: Exploring Geographical Variation. *Environmental Health Perspectives* 111(1):39–44. PMID: 12515677; PMCID: PMC1241304; doi: 10.1289/ehp.5181.

18. Johnson E, **Dominici F**, Griswold M, Zeger SL (2003). Disease Cases and Their Medical Costs Attributable to Smoking: An Analysis of the National Medical Expenditure Survey. *Journal of Econometrics* 112(1):135–151.

19. Samet JM, **Dominici F**, McDermott A, Zeger SL (2003). New Problems for an Old Design: Time-Series Analyses of Air Pollution and Health. *Epidemiology* 14(1):11–12.

20. Burnett R, Dewanji A, **Dominici F**, Cohen A, Goldberg M, Krewski D (2003). On the Relationship Between Time Series Studies, Dynamic Population Studies, and Estimating Life Lost Due to Short-Term Exposure to Environmental Risks. *Environmental Health Perspectives* 111:1170–1174. PMCID: PMC1241570.

21. **Dominici F**, McDermott A, Samet JM, Zeger SL (2003). Airborne Particulate Matter and Mortality: Time-Scale Effects in Four US Cities (with invited commentary). *American Journal of Epidemiology* 157:1055–1065.

22. **Dominici F**, McDermott A, Samet JM, Zeger SL (2003). Response to Dr. Smith: Timescale-Dependent Mortality Effects of Air Pollution. *American Journal of Epidemiology* 157:1071–1073.

23. **Dominici F**, Burnett R (2003). Risk Models for Particulate Air Pollution. *Journal of Toxicology and Environmental Health* 66(16–19):1883–1890.

24. **Dominici F**, Sheppard L, Clyde M (2003). Health Effects of Air Pollution: A Statistical Review. *International Statistical Review* 71(2):243–276.

25. Daniels M, **Dominici F**, Zeger SL (2004). Underestimation of Standard Errors in Multi-Site Time-Series Studies of Air Pollution and Health. *Epidemiology* 15:57–62.

26. Daniels MJ, **Dominici F**, Zeger SL, Samet JM (2004). The National Morbidity, Mortality and Air Pollution Study, Part III: PM10 concentration-response curves and thresholds for the 20 largest US cities. *Health Effects Institute Research Report*, 94(3):1–21. (discussion 23–30)

27. **Dominici F** (2004). Time-Series Analysis of Air Pollution and Mortality: A Statistical Review. *Health Effects Institute Research Report*, 123:3–27. (discussion 29–33)

28. Bell M, Samet JM, **Dominici F** (2004). Time-Series Studies of Particulate Matter. *Annual Review of Public Health*, 25:247–280.

29. Jackson L, Correa A, Lees PSJ, **Dominici F**, Stewart P, Breysse PA, Matanoski G (2004). Parental Lead Exposure and Total Anomalous Pulmonary Venous Return. *Birth Defects Research*, 70:185–193.

30. Merialdi M, Caulfield L, Zavaleta N, Figueroa A, Costigan KM, **Dominici F**, DiPietro J (2004). Randomized Controlled Trial of Prenatal Zinc Supplementation and Fetal Growth. *American Journal of Clinical Nutrition*, 79:826–830.

31. Merialdi M, Caulfield L, Zavaleta N, Figueroa A, Costigan KM, **Dominici F**, DiPietro J (2004). Randomized Controlled Trial of Prenatal Zinc Supplementation and the Development of Fetal Heart Rate Patterns. *American Journal of Obstetrics and Gynecology*, 190:1106–1112.

32. **Dominici F**, Zanobetti A, Zeger SL, Schwartz J, Samet JM (2004). Hierarchical Bivariate Time-Series Models: A Combined Analysis of the Effects of Particulate Matter on Morbidity and Mortality. *Biostatistics*, 5:341–360.

33. **Dominici F**, Cole S, Spiegelman D (2004). Methodological contributions to the American Journal of Epidemiology. *American Journal of Epidemiology*, 160:197–198.

34. Yurk RA, Diette GB, Skinner EA, **Dominici F**, Clark RD, Steinwachs DM, Wu A (2004). Predicting Patients Reported Asthma Outcomes for Adults in Managed Cares. *American Journal of Managed Care*, 11:322–328.

35. Bell M, Samet JM, McDermott A, Zeger SL, **Dominici F** (2004). Ozone and Mortality in 95 U.S. Urban Communities from 1987 to 2000. *Journal of the American Medical Association*, 292:2372–2378.

36. **Dominici F**, McDermott A, Hastie T (2004). Improved Semi-parametric Time Series Models of Air Pollution and Mortality. *Journal of the American Statistical Association*, 468:938–948.

37. Basu R, **Dominici F**, Samet JM (2005). Temperature and Mortality Among the Elderly in the United States: A Comparison of Epidemiologic Methods. *Epidemiology*, 16(1):58–66.

38. Huang IC, Frangakis C, **Dominici F**, Diette GB, Wu A (2005). Application of a Propensity Score Approach for Risk Adjustment in Multiple Physician Group Profiling on Asthma Care. *Health Services Research*, 40:253–278. PMID: 15663712; PMCID: PMC1361136; doi:10.1111/j.1475-6773.2005.00352.x.

39. Huang IC, **Dominici F**, Frangakis C, Diette GB, Damberg CL, Wu A (2005). Is Risk-Adjustor Selection More Important than Statistical Approach for Provider Profiling? Asthma as an Example. *Medical Decision Making*, 25:20–359.

40. Huang IC, Diette GB, **Dominici F**, Frangakis C, Wu A (2005). Variations of Physician Group Profiling Indicators. *American Journal of Managed Care*, 11:38–49.

41. Peng R, **Dominici F**, Pastor-Barriuso R, Zeger SL, Samet JM (2005). Seasonal Analyses of Air Pollution and Mortality in 100 U.S. Cities. *American Journal of Epidemiology*, 161:585–594.

42. Merialdi M, Caulfield LE, Zavaleta N, Figueroa A, Costigan KA, **Dominici F**, DiPietro JA (2005). Randomized, Controlled Trial of Prenatal Zinc Supplementation and Fetal Bone Growth. *Obstetrical and Gynecological Survey*, 60:13–15.

43. Merialdi M, Caulfield LE, Zavaleta N, Figueroa A, Costigan KA, **Dominici F**, DiPietro JA (2005). Fetal Growth in Peru: Comparisons with International Fetal Size Charts and Implications for Fetal Growth Assessment. *Ultrasound in Obstetrics and Gynecology*, 2:123–128.

44. **Dominici F**, Levy J, Louis TL (2005). Methodological Challenges and Contributions in Disaster Epidemiology. *Epidemiologic Reviews*, 27:1–4.

45. Bell M, **Dominici F**, Samet JM (2005). A Meta-Analysis of Time-Series Studies of Ozone and Mortality with Comparison to the National Morbidity Mortality Air Pollution Study (with discussion). *Epidemiology*, 16:436–445. PMID: 15951661; PMCID: PMC3581312;

doi:10.1097/01.ede.0000165817.40152.85.

46. Huang Y, **Dominici F**, Bell M (2005). Bayesian Hierarchical Distributed Lag Models for Summer Ozone Exposure and Cardio-Respiratory Mortality. *Environmetrics*, 16:547–562. PMID: 23825932; PMCID: PMC3697867; doi:10.1002/env.721.

47. Fung KY, Krewski D, Burnett R, **Dominici F** (2005). Testing the Harvesting Hypothesis. *Journal of Toxicology and Environmental Health*, 68:1137–1154.

48. Eftim S, **Dominici F** (2005). Multi-site Time-Series Studies versus Cohort Studies: Methods, Findings, and Policy Implications. *Journal of Toxicology and Environmental Health*, 68:1191–1205.

49. **Dominici F**, McDermott A, Daniels M, Zeger SL, Samet JM (2005). Revised Analysis of the National Morbidity, Mortality and Air Pollution Study: Mortality Among Residents of 90 Cities. *Journal of Toxicology and Environmental Health*, 68:1071–1092.

50. **Dominici F**, Cope L, Naiman D, Zeger SL (2005). Smooth Quantile Ratio Estimation (SQUARE). *Biometrika*, 92:543–557.

51. Choi L, **Dominici F**, Zeger SL (2005). Estimating Treatment Efficacy Over Time: a Logistic Regression Model for Binary Longitudinal Outcomes. *Statistics in Medicine*, 24:2789–2805.

52. **Dominici F**, Zeger SL (2005). Smooth Quantile Ratio Estimation with Regression: Estimating Medical Expenditures for Smoking Attributable Diseases. *Biostatistics*, 6:505–519.

53. **Dominici F**, Zeger SL, Parmigiani G, Katz J, Christian P (2006). Estimating Percentile-Specific Effects in Counterfactual Models: A Case Study of Micronutrient Supplementation, Birth Weight, and Infant Mortality. *Journal of the Royal Statistical Society, Series C*, 55(2):261–280.

54. Peng R, **Dominici F**, Louis T (2006). Model Choice in Multi-Site Time Series Studies of Air Pollution and Mortality. *Journal of the Royal Statistical Society, Series A*, 169(2):179–203.

55. Buja A, Mastrangelo G, Perissinotto E, Rausa G, Canova C, Grigoletto F, **Dominici F** (2006). Cancer Incidence among Female Cabin Attendants: A Meta-Analysis of Published Data. *Journal of Women's Health*, 15:98–105.

56. Sapkota A, Halden R, **Dominici F**, Groopman J, Buckley T (2006). Urinary Biomarkers of 1,3-Butadiene in Environmental Settings Using Liquid Chromatography Isotope Dilution Tandem Mass Spectrometry. *Chemico-Biological Interactions*, 160:70–79.

57. **Dominici F**, Peng D, Bell M, Pham M, McDermott A, Zeger SL, Samet JM (2006). Fine Particulates Air Pollution and Hospital Admission for Cardiovascular and Respiratory Diseases. *Journal of the American Medical Association*, 295:1127–1135. PMID: 16522832; PMCID: PMC3543154; doi:10.1001/jama.295.10.1127.

58. Katz J, Christian P, **Dominici F**, Zeger SL (2006). Treatment Effects of Maternal Micronutrient Supplementation vary by Percentiles of the Birth Weight Distribution in rural Nepal. *Journal of Nutrition*, 136:1389–1394.

59. Bell M, Peng R, **Dominici F** (2006). The Exposure-Response Curve for Ozone and Risk of Mortality and the Adequacy of Current Ozone Regulations. *Environmental Health Perspectives*, 114(4):532–536. PMID: 16581541; PMCID: PMC1440776; doi:10.1289/ehp.8816.

60. Peng R, **Dominici F**, Zeger SL (2006). Reproducible Epidemiological Research. *American Journal of Epidemiology*, 163:783–789.

61. Symons JM, Wang L, Guallar E, Howell E, **Dominici F**, Schwab M, Ange BA, Samet JM, Ondov J, Harrison D, Geyh A (2006). A Case-Crossover Study of Fine Particulate Matter Air Pollution and Congestive Heart Failure Hospitalization. *American Journal of Epidemiology*, 164:421–433.

62. Thomas D, Jerrett M, Kuenzli N, Louis T, **Dominici F**, Zeger SL, Schwartz J, Burnett R, Krewski D, Bates D (2007). Bayesian Model Averaging in Time-Series Studies of Air Pollution and Mortality. *Journal of Toxicology and Environmental Health*, 70(3–4):311–315.

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20. **Dominici F** (June 2026). AI: Solution or obstacle for healthy adaptation to extreme weather events? [lecture: National Institutes of Health WALS lecture series 2026-2027].

21. **Dominici F** (August 2026). AI: Solution or obstacle for climate adaptation? [keynote: ISEE Dan Greenbaum lecture, International Society for Environmental Epidemiology Annual Conference].

22. **Dominici F** (September 2026). AI: Solution or obstacle for healthy adaptation to extreme weather events? [lecture: Florida International University].

23. Vartziotis T, Kosteli R, Vartziotis ED, Dasoulas G, Keckeisen M, Skianis K, Kotsopoulos S, **Dominici F** (2026). From tokens to watt-hours: analytical energy estimation for LLM inference on modern GPUs. GREEN-AI Workshop at ECML-PKDD 2026. Accepted for oral presentation.

24. Kleutgens J, Tec M, Battiloro C, **Dominici F**, Daras G (2026). The wrong data at the right time: off-domain mixing for discrete diffusion. BeNTo Workshop, NeurIPS 2026. OpenReview [Internet]. Published 2026 Sep 28.

    Available from: https://openreview.net/forum?id=HM3OejMCjy

25. Kleutgens J, Tec M, Battiloro C, **Dominici F**, Daras G (2026). Ambient discrete diffusion: using the wrong data at the right time for data efficient learning. arXiv [Internet]. 2026 Oct 8 arXiv:2610.12340. doi:10.48550/arXiv.2610.12340.

    Available from: https://arxiv.org/pdf/2610.12340

GRANTS

1. P50MD010428 (Laden, Levy) 07/2015 – 06/2020. NIH/NIMHD. Disparities in Exposure and Health Effects of Multiple Environmental Stressors Across the Life Course. Role: Co-I.

2. 83615601 (Laden) 07/2015 – 07/2020. EPA. Disparities in Exposure and Health Effects of Multiple Environmental Stressors Across the Life Course. Role: Co-I.

3. 83587201-0 (Koutrakis) 12/2015 – 12/2020. EPA. Regional Air Pollution Mixtures: The Past and Future Impacts of Emission Controls and Climate Change on Air Quality and Health. Role: Co-I.

4. 4953-RFA14-3/16-4 (**Dominici**, Zanobetti) 02/2016 – 02/2020. HEI. Assessing Adverse Health Effects of Long-Term Exposure to Low Levels of Ambient Pollution. Role: PI.

5. R01ES026217 (Zigler) 02/2016 – 02/2021. NIH. Causal Inference with Interference for Evaluating Air Quality Policies. Role: Co-I.

6. R01MD012769 (Bell) 09/2017 – 06/2022. NIH. Environmental Health Disparities in an Older Population. Role: Consortium PI.

7. DP2MD012722 (Rose) 09/2017 – 07/2022. NIH. Machine Learning for Health Outcomes and Quality of Care in Low-Income Populations. Role: Co-I.

8. R01ES028033 (Laden, **Dominici**) 12/2017 – 12/2022. NIH. Relationship Between Multiple Environmental Exposures and CVD Incidence and Survival: Vulnerability and Susceptibility. Role: PI.

9. Project No: NA (**Dominici**) 07/2018 – 06/2024. Fernholz Foundation. Advancing Data Science Research at the Harvard T.H. Chan School of Public Health. Role: PI.

10. 5R01AG060232 (Setoguchi-Iwata) 06/2019 – 04/2024. NIH. Disease Outcomes iN Older adults under extreme Heat, AiR pollution and Medication use (DO-NO-HARM). Role: Consortium PI.

11. 1R01ES030616 (Kioumourtzoglou) 03/2020 – 12/2024. NIH. Integrating Air Pollution Prediction Models: Uncertainty Quantification and Propagation in Health Studies. Role: Consortium PI.

12. 1R01AG066793 (Zanobetti) 04/2020 – 11/2025. NIH. National Cohort Studies of Alzheimer's Disease, Related Dementias and Air Pollution. Role: Co-I.

13. 1R01ES029950 (Wellenius) 05/2020 – 10/2023. NIH. Evidence to Improve Heat Warning Effectiveness in Reducing Morbidity and Mortality. Role: Co-I.

14. 2020 Star Friedman Award (**Dominici**) 07/2020 – 07/2021. Harvard University. Air Pollution, Race, and Health Outcomes for COVID-19 in the United States: Data Access, Methods, and Dissemination. Role: PI.

15. G-2020-13946 (**Dominici**) 09/2020 – 08/2023. Alfred P. Sloan Foundation. Causal Inference with Complex Treatment Regimes: Design, Identification, Estimation, and Heterogeneity. Role: PI.

16. P30ES000002 (Weisskopf) 11/2020 – 11/2021. NIH/NIEHS. Air Pollution, Race, and Health Outcomes for COVID-19 in the United States: Data Science Methods and Dissemination. Role: PI.

17. 1RF1AG074372-01A1 (Patel, **Dominici**, Estiri) 06/2022 – 05/2025. NIH. The Confluence of Extreme Heat/Cold on the Health and Longevity of an Aging Population with Alzheimer's and Related Dementia. Role: Co-PI.

18. 1R01MD016054-01A1 (Bell, Deziel) 07/2022 – 03/2027. NIH. Air Pollution, Heat, Cold, and Health: Disparities in the Rural South. Role: PD/PI.

19. 1R01ES034373-01 (Zanobetti, Braun) 09/2022 – 07/2027. NIH. Characterizing the Link between Multiple Environmental Exposures and Parkinson's Disease Exacerbation. Role: MPI.

20. Project No: NA (**Dominici**) 10/2022 – 09/2023. Harvard University. Data Science Methods and Tools for Climate Change Solutions. Role: PI.

21. 1RF1AG080948 (Bell, Gill, Benmarhnia) 12/2022 – 11/2025. NIH. Susceptibility and Adverse Health Outcomes Related to Climate-Sensitive Events among Older Medicare Beneficiaries with Alzheimer's and Dementia. Role: Co-PI.

22. 1U24ES035309 (Wellenius, **Dominici**, Nori-Sarma) 05/2023 – 04/2026. NIH. CAFÉ: A Research Coordinating Center to Convene, Accelerate, Foster, and Expand the Climate Change and Health Community of Practice. Role: MPI.

23. R01ES34021 (Ertefaie, **Dominici**, Strawderman) 09/2023 – 08/2026. NIH. Statistical Methods to Characterize Causal Mechanisms by which Air Pollution Affects the Recurrence of Cardiovascular Events. Role: MPI.

24. Project No: NA (**Dominici**) 05/2024 – 04/2027. Amazon Web Services. AWS Research Project. Role: PI.

25. R01ES037156 (**Dominici**) 07/2024 – 06/2028. NIH. Synergizing Topological Deep Learning and Spatio-Temporal Causal Inference to Unpack Health Disparities in the Climate Crisis. Role: PI.

26. 2418829 (Rosenman) 07/2024 – 06/2026. NSF. LEAPS-MPS: Empirical Bayes Evidence Synthesis Techniques for Hybridizing Observational and Experimental Data. Role: PI.

27. Training Grant in Environmental Biostatistics. Role: Co-Director (Brent Coull, PI)

TEACHING

Advisees at Johns Hopkins

Pre-doctoral students in Biostatistics (Primary advisor, Hopkins)

- Sorina Eftim, PhD candidate, 2003-2006, completed
- Sergio Venturini, PhD candidate, Visiting Scholar from Bocconi University, Italy *Awarded in 2008 as the best Italian PhD Thesis*
- Yijie Zhou, PhD candidate, 2004-2007, completed
- Howard Chang, PhD candidate, 2006-2010, completed *ENAR 2008 Student Travel Award*
- Jessica Myer, PhD candidate, 2007-2010, completed *ENAR 2008 Student Travel Award*
- Jennifer Bobb, PhD candidate, 2008-2010, completed *ENAR 2010 Student Travel Award*

Post-doctoral fellows in Biostatistics (Primary advisor, Hopkins)

- Michelle Bell, PhD 2002-2003, Professor, Yale University
- Roger Peng, PhD, 2003-2005, Associate Professor, Johns Hopkins University
- Holly Janes, PhD, 2005-2007, Assistant Professor, University of Washington Seattle
- Sonja Greven, PhD, 2008-2009, Assistant Professor, University of Munich
- Christopher D. Barr, PhD, 2008-2010, Assistant Professor, Harvard University

Statistical Advisor of PhD Students from Other Departments (Hopkins)

- Mario Merialdi, PhD International Health, 2001
- Rupa Basu, PhD Epidemiology, 2002
- I-Chan Huang, PhD Health Policy and Management, 2003
- Leila Jackson, PhD Epidemiology, 2003
- Leslie Cope, PhD Mathematical Sciences, 2003
- Morel Symons, PhD Epidemiology, 2005

- Sung Roul Kim, PhD Environmental Health Science, 2006
- Hari Nathan, MD, Clinical Research Scholar Programs, Department of Surgery, 2007
- Suzanna Nazarian, MD, Clinical Research Scholar Programs, Department of Surgery, 2007

MPH Students (Hopkins)

- Jayme Locke, MD, Resident, Department of Surgery, Johns Hopkins Hospital, 2006 *MPH Capstone Award Winner*
- Lia Clattenburg, MD, School of Medicine, Johns Hopkins Hospital, 2006
- Jose Rodriguez-Paz, MD, Department of Anesthesiology, Johns Hopkins Hospital, 2007
- Robert Meguid, MD, Department of Surgery, Johns Hopkins Hospital, 2007
- Jeff Schoenborn, BS, Battelle Memorial Institute, 2008

Final Oral Participation (Hopkins)

- Karlyn Heiner, ScM Biostatistics, 1998
- Huey-Fang Sun, PhD Mental Hygiene, 2000
- Paul Muntner, PhD Epidemiology, 2000
- Rupa Basu, PhD Epidemiology, 2001
- Mario Merialdi, PhD International Health, 2001
- Denys Lau, PhD Health Policy and Management, 2002
- I-Chan Huang, PhD Health Policy and Management, 2003
- Leila Jackson, PhD Epidemiology, 2003
- Leslie Cope, PhD Mathematical Sciences, 2003
- Samuel Mills, PhD Population and Family Health Science, 2004
- Amir Sapkota, PhD Environmental Health Science, 2005
- Morel Symons, PhD Epidemiology, 2005
- Sung Roul Kim, PhD Environmental Health Science, 2006
- Yingkai Cheng, PhD Epidemiology, 2006
- Arantxa Conchero, PhD Epidemiology, 2007
- Jennifer Taylor, PhD Health Policy and Management, 2007
- Yijie Zhou, PhD Biostatistics, 2007
- Nicole Cardello, PhD Environmental Health Science, 2007
- Michael Johansson, PhD Molecular Biology and Immunology, 2008

- Lilly Engineer, PhD Health Policy and Management, 2008
- Kenny Shum, PhD Biostatistics, 2008
- Juan Ramos-Bonilla, PhD Environmental Health Sciences, 2009
- Howard Chang, PhD Biostatistics, 2009
- Lydia Isaac, PhD Health Policy and Management, 2009

Advisees at Harvard

Undergraduate Students in Statistics (Primary Advisor)

- Uggi Lee, 2012-2013
- Barrett Jansen, 2014-2015
- Kaela Nelson, 2018-2019
- Tara Ippolito, 2018-2019
- Sophie Woodward, 2022
- Michelle Qin, 2022-2023
- Daniela Garcia, 2023-2024
- Leo Vanciu, 2023-present
- Sophie-An Kingsbury-Lee, 2023-present

Pre-Doctoral Students in Biostatistics (Primary Advisor)

- Andrew Correia, PhD candidate, 2010-2013
- Krista Watts, PhD candidate, 2011-2013
- Matt Cefalu, PhD candidate, 2011-2013
- Joseph Antonelli, PhD candidate, 2013-2017
- Elisabeth Smoot, PhD candidate, 2014-2015
- Tai Cai, PhD candidate, 2014-2017
- Yeting Du, PhD candidate, 2016-2017 (not primary)
- Georgia Papadogeorgou, PhD candidate, 2014-2018
- Emma Thomas, PhD candidate, 2017-2020
- Shirley Liao, PhD candidate, 2018-2019
- Xiao Wu, PhD candidate, 2017-2021
- Irina Degtiar, PhD candidate, 2017-2021

- Yue Yang, PhD candidate, 2020-Present
- Jenny Lee, PhD candidate, 2020-2025
- Ellen Considine, PhD candidate, 2021-2025
- Michael Cork, PhD candidate, 2021-2025
- Davide Bray, PhD, 2022-2026
- Sophie Woodward, 2022-Present
- Suhwan Bong, 2023-Present
- James Kitch, 2025-Present

Post-Doctoral Fellows in Biostatistics (Primary Advisor)

- Yeonsung Chang, PhD, 2010-2011, Assistant Professor of Statistics, Seoul University
- Yiping Dou, PhD, 2010, Scientist at Yahoo
- David Diez, PhD, 2010-2012, Google
- Lauren Hund, PhD, 2012-2013, Statistician, Sandia National Laboratories
- Dhruv Sharma, PhD, 2011-2012, Michigan State University
- Corwin Zigler, PhD, 2010-2013, Associate Professor of Statistics, University of Texas at Austin
- Matt Cefalu, PhD, 2013-2014, Rand
- Jennifer F Bobb, PhD, 2011-2015, Associate Professor of Biostatistics, Group Health Research Institute
- Fei Jiang, PhD, 2013-2015, Assistant Professor of Statistics, University of South Carolina
- Fred Cudhea, PhD, 2013-2015, Research Scientist, Tufts University
- Kyu Ha Lee, PhD, 2011-2015, Assistant Professor of Nutrition, Harvard T.H. Chan School of Public Health
- Danielle Braun, PhD, 2013-2017, Senior Scientist, Biostatistics, Harvard T.H. Chan School of Public Health
- Ander Wilson, PhD, 2014-2017, Assistant Professor of Statistics, Colorado State University
- Chanmin Kim, PhD, 2014-2018, Assistant Professor of Statistics, Sungkyunkwan University, South Korea
- Joseph Antonelli, PhD, 2015-2018, Assistant Professor of Statistics, University of Florida
- Aaron Fisher, PhD, 2016-2019, Data Scientist, Pfizer
- Georgia Papadogeorgou, PhD, 2017-2019, Assistant Professor of Statistics, University of Florida

- Kwonsang Lee, PhD, 2017-2019, Assistant Professor of Statistics, Sungkyunkwan University, South Korea
- Rachel Nethery, PhD, 2017-2019, Assistant Professor of Biostatistics, Harvard T.H. Chan School of Public Health
- Boyu Ren, PhD, 2017-2020, Assistant Professor of Biostatistics, McLean Hospital
- Xiao Wu, PhD, 2021-2023, Assistant Professor, Columbia Mailman School of Public Health
- Falco Bargagli Stoffi, PhD, 2020-2023, Assistant Professor, University of California Los Angeles
- Kevin Josey, PhD, 2020-2023, Assistant Professor, School of Public Health, University of Colorado
- Kate Hu, PhD, 2021-2024, Assistant Professor, Department of Statistics, Ohio State University
- Daniel Mork, PhD, 2021-2024, Research Scientist, Harvard T.H. Chan School of Public Health
- Mauricio Tec, PhD, 2022-2025, Senior Applied Scientist, AGI, Amazon
- Michael Cork, PhD, 2026, Consultant, Empower Analytics
- Heejun Shin, PhD, 2024-2026 item Claudio Battiloro, PhD, 2024-2026, Assistant Professor, Brown University
- Gianluca Guidi, PhD, 2024-Present
- Federica Spoto, PhD, 2024-Present
- Xinran Liu, PhD, 2026-Present
- Davide Bray, PhD, 2026-Present

Statistical Advisor of PhD students from Other Departments

- Kathie Dionisio, PhD Department of Global Health and Population, 2010
- Peter James, PhD Environmental Health, 2011
- Marianthi-Anna Kioumourtzoglou, PhD Environmental Health, 2013
- Mercedes Bravo, PhD Yale School of Forestry and Environmental Studies, 2013
- Memo-Cedeno Laurent, PhD Environmental Health, 2014
- Yan Wang, PhD Environmental Health, 2017
- Qian Di, PhD Environmental Health, 2017
- Jong Eun Rhee, PhD Environmental Health, 2017

Other

- Jason Wasfy, MD, MPhil, Massachusetts General Hospital, 2016 *Recipient of KL2/Catalyst Medical Research Investigator Training Award*

Final Oral Participation (PhD)

- Kathie Dionisio, Environmental Health, 2011
- Mariel Finucane, Biostatistics, 2011
- Peter James, Epidemiology, 2012
- Richard White, Biostatistics, 2012
- Marianthi-Anna Kioumourtzoglou, Environmental Health, 2013
- Mathieu Maheu-Girox, Global Health and Population, 2013
- Frederick Cudhea, Biostatistics, 2013
- Memo-Cedeno Laurent, Environmental Health, 2014
- Mathieu Maheu-Giroux, Global Health Population, 2014
- Elisabeth Smoot, Biostatistics, 2015
- Joseph Antonelli, Biostatistics, 2015

K99 Advisees

- Jason Wasfy, MD, MPhil, Massachusetts General Hospital, 2015
- Xabier Adrian Garcia-De-Albeniz, MD, PhD, Harvard T.H. Chan School of Public Health, 2016
- Peter James, ScD, Harvard T.H. Chan School of Public Health, 2016, 2021
- Gitanjali Singh, PhD, Tufts University Friedman School of Nutrition Science and Policy, 2016
- Mercedes Bravo, PhD, University of Michigan School of Natural Resources and Environment, 2016
- Brooke Anderson, PhD, University of Colorado, 2017
- Kelvin Fong, PhD, Yale University School of Forestry and Environmental Studies, 2020, 2021
- Seulkee Heo, PhD, Yale University School of Forestry and Environmental Studies, 2020
- Rachel Nethery, PhD, Harvard T.H. Chan School of Public Health, 2020, 2021
- Briana Stephenson, PhD, Harvard T.H. Chan School of Public Health, 2020, 2021
- Whanhee Lee, PhD, Yale University School of the Environment, 2021
- Hannah Correia, PhD, Harvard T.H. Chan School of Public Health, 2021
- Isabel Fulcher, MD, Harvard Medical School, 2021
- Robbie Parks, PhD, Columbia University, 2021
- Ashley Buchanan, DrPH, Harvard T.H. Chan School of Public Health, 2021
- Longxiang Li, PhD, DrPH, Harvard T.H. Chan School of Public Health, 2022
- Yanguang Wei, PhD, DrPH, Harvard T.H. Chan School of Public Health, 2022
- Xiao Wu, PhD, Mailman School of Public Health, Columbia University, 2024